Cross-step interference, directional curvature, and gradient descent

1 Introduction

Training large language models (LLMs) relies on adaptive optimization, momentum, gradient clipping, learning-rate schedules and other heuristics to stabilize learning and mitigate phenomena such as loss spikes and oscillatory updates. Yet, despite their effectiveness, it is not fully understood why these mechanisms work, and under what conditions they fail.

To address this gap, we identify a persistent behaviour of transition between two distinct learning regimes. Early in training, successive gradients tend to align, allowing updates to accumulate constructively as the loss decreases. Later, gradient directions reverse more frequently, causing successive updates to partially cancel one another. Despite this increased cancellation, the loss can continue to decrease.

Two complementary intuitions help explain this behavior. From the perspective of representation learning, each new mini-batch updates parameters that also encode previously learned structure. Depending on its direction, an update may reinforce existing representations or partially undo them. From the perspective of optimization geometry, a step may overshoot the bottom of a loss valley, prompting subsequent steps to reverse direction and partially retrace the path. Both mechanisms can give rise to oscillatory parameter updates even as optimization continues to make progress.

We formalize these dynamics using identities for stochastic gradient descent (SGD), relating the alignment of consecutive gradients to curvature and gradient noise, and cumulative update cancellation to net parameter displacement. Building on these relations, we introduce scalable diagnostics for quantifying these dynamics and tracking the transition between constructive and oscillatory regimes during large-scale model training.

2 Parameter displacement under stochastic gradient descent

Let $\theta_t \in \mathbb{R}^p$ denote the parameters after update t, and let g_t denote a mini-batch gradient evaluated at θ_{t-1}. We consider constant-step SGD with learning rate $\eta > 0$, without momentum, weight decay, or clipping

$$\theta_t = \theta_{t-1} - \eta g_t. \tag{1}$$

After T updates, summing Eq. (1) gives the displacement from initialization,

$$\Delta_T := \theta_T - \theta_0 = -\eta \sum_{t=1}^{T} g_t. \tag{2}$$

Expanding its squared norm separates individual gradient magnitudes from their temporal interactions

$$\boxed{\|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 + 2\eta^2 \sum_{1 \le u < t \le T} g_u^\top g_t.} \tag{3}$$

The gradient-energy terms $\|g_t\|^2$ are nonnegative. Each pairwise inner product $g_u^\top g_t$ can be positive, zero, or negative: aligned gradients reinforce their contribution to the sum, whereas opposing gradients cancel part of it. Gradient energy is not itself net displacement; the interaction term determines how much of this energy remains in the squared norm of the accumulated vector.

For nonzero consecutive gradients, define their cosine similarity as

$$\rho_t := \frac{g_{t-1}^\top g_t}{\|g_{t-1}\| \, \|g_t\|}, \qquad t = 2, \dots, T. \tag{4}$$

Separating adjacent pairs from the remaining interactions yields

$$\begin{aligned} \|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \underbrace{\sum_{t=2}^{T} \|g_{t-1}\| \, \|g_t\| \, \rho_t}_{\text{lag-one interaction}} \\ + 2\eta^2 \underbrace{\sum_{t=3}^{T} \sum_{u=1}^{t-2} g_u^\top g_t}_{\text{interaction beyond lag one}} . \end{aligned} \tag{5}$$

Negative ρ_t contributes cancellation between consecutive updates. However, longer-lag interactions can reinforce or offset these negative contributions. Thus, adjacent anti-alignment diagnoses a local interaction, while the complete double sum determines longer-term cumulative interaction.

3 Curvature, gradient noise, and consecutive alignment

3.1 A local expansion along the SGD trajectory

Let $\mathcal{L}$ denote the full training objective. We write a stochastic gradient as

$$g_t = \nabla \mathcal{L}(\theta_{t-1}) + \xi_t, \tag{6}$$

where ξ_t is the gradient estimation error. Suppose $\mathcal{L}$ is twice continuously differentiable along the step under consideration, and define $H_{t-1} := \nabla^2 \mathcal{L}(\theta_{t-2})$. Expanding the full gradient around θ_{t-2} and substituting the preceding SGD step gives

$$g_t = (I - \eta H_{t-1}) g_{t-1} + (\xi_t - \xi_{t-1}) + r_t, \tag{7}$$

where r_t is the Taylor remainder. Neglecting r_t recovers the first-order approximation. Taking the inner product with g_{t-1} then gives

$$\rho_t \approx \frac{\|g_{t-1}\|^2 - \eta g_{t-1}^\top H_{t-1} g_{t-1} + g_{t-1}^\top (\xi_t - \xi_{t-1})}{\|g_{t-1}\| \, \|g_t\|}. \tag{8}$$

The numerator contains the preceding gradient energy, a curvature contribution, and the change in gradient noise. Positive curvature along the preceding gradient direction reduces its alignment with the next gradient. The noise difference can reinforce or oppose this effect on an individual step.

We normalize the curvature and noise terms by the preceding gradient energy:

$$\kappa_t := \frac{g_{t-1}^\top H_{t-1} g_{t-1}}{\|g_{t-1}\|^2}, \qquad \varepsilon_t := \frac{g_{t-1}^\top (\xi_t - \xi_{t-1})}{\|g_{t-1}\|^2}. \tag{9}$$

With these definitions, Eq. (8) becomes

$$\boxed{\rho_t \approx \frac{\|g_{t-1}\|}{\|g_t\|}(1 - \eta\kappa_t + \varepsilon_t).} \tag{10}$$

The norm ratio is positive, so the first-order sign prediction depends on $1 - \eta\kappa_t + \varepsilon_t$ (Table 1). A larger positive directional curvature favors reversal; noise shifts the corresponding boundary. These conditions describe alignment, not the separate question of whether an oscillation is damped or unstable.

Table 1: Consecutive-gradient alignment predicted by the first-order model. The Taylor remainder is neglected; these are not exact sign equivalences for an arbitrary nonlinear step.

Alignment	Cosine sign	First-order condition
Positive	$\rho_t > 0$	$\eta\kappa_t < 1 + \varepsilon_t$
Orthogonal	$\rho_t = 0$	$\eta\kappa_t = 1 + \varepsilon_t$
Negative	$\rho_t < 0$	$\eta\kappa_t > 1 + \varepsilon_t$

The remainder matters near this boundary. In fact, Eq. (7) implies the exact relation

$$\rho_t = \frac{\|g_{t-1}\|}{\|g_t\|}(1 - \eta\kappa_t + \varepsilon_t + \delta_t), \qquad \delta_t := \frac{g_{t-1}^\top r_t}{\|g_{t-1}\|^2}. \tag{11}$$

The first-order sign is reliable only to the extent that this omitted term does not change the numerator's sign.

3.2 Directional curvature and a gradient-derived secant

The quantity κ_t is a Hessian Rayleigh quotient along the preceding stochastic gradient. If $H_{t-1} v_i = \lambda_i v_i$ in an orthonormal eigenbasis and $g_{t-1} = \sum_i a_i v_i$, then

$$\kappa_t = \frac{\sum_i \lambda_i a_i^2}{\sum_i a_i^2}. \tag{12}$$

Thus, the gradient selects an eigenvalue-weighted average rather than the maximum eigenvalue. A sharp direction can exist without contributing much to the current gradient. This distinction is important when comparing alignment with the largest Hessian eigenvalue.

Rearranging Eq. (10) motivates the observable quantity

$$\eta\widehat{\kappa}_t := 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|} + \varepsilon_t. \tag{13}$$

Equation (11) shows that $\widehat{\kappa}_t = \kappa_t - \delta_t/\eta$. Consequently, it approximates the pointwise directional curvature when the Taylor remainder is small. For mini-batch gradients, the noise term requires additional information and cannot be recovered from adjacent cosine and norms alone.

For full-batch GD, $\xi_t = 0$ and hence $\varepsilon_t = 0$. The recorded gradient-only quantity is then exactly

$$\eta\widehat{\kappa}_t = 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|} = -\frac{g_{t-1}^\top (g_t - g_{t-1})}{\|g_{t-1}\|^2}. \tag{14}$$

It measures the change in the gradient projected onto its preceding direction. In exact-arithmetic GD, the fundamental theorem of calculus provides a finite-step curvature interpretation:

$$\widehat{\kappa}_t = \frac{g_{t-1}^\top \overline{H}_t g_{t-1}}{\|g_{t-1}\|^2}, \qquad \overline{H}_t = \int_0^1 \nabla^2 \mathcal{L}(\theta_{t-2} - s\eta g_{t-1})\, ds. \tag{15}$$

The secant therefore samples curvature averaged along the preceding step. It equals the pointwise Rayleigh quotient on a quadratic objective, but need not equal either the local maximum eigenvalue or its finite-step average on a nonlinear trajectory.

3.3 The noise contribution under unbiased sampling

To isolate the stochastic term, assume fresh, conditionally unbiased mini-batch sampling with finite second moments. Let $\mathcal{F}_{t-1}$ contain the history through the preceding update, so that $\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$. Since g_{t-1} is measurable with respect to this history,

$$\mathbb{E}[g_{t-1}^\top \xi_t] = 0. \tag{16}$$

The preceding gradient already contains ξ_{t-1}. Conditional unbiasedness at that earlier step removes its cross term with $\nabla\mathcal{L}(\theta_{t-2})$, giving

$$\mathbb{E}[g_{t-1}^\top \xi_{t-1}] = \mathbb{E}[\|\xi_{t-1}\|^2]. \tag{17}$$

Subtracting these two expectations yields

$$\boxed{\mathbb{E}[g_{t-1}^\top (\xi_t - \xi_{t-1})] = -\mathbb{E}[\|\xi_{t-1}\|^2] \le 0.} \tag{18}$$

Even unbiased noise therefore produces a nonpositive expected contribution to the unnormalized adjacent inner product in this decomposition. This identifies a mechanism that can reduce successive-gradient alignment without requiring large positive directional curvature. It is not a statement that the expected cosine must be negative: the total numerator has other contributions and the cosine denominator is random. Correlated sampling or sampling without the stated conditional unbiasedness need not satisfy this identity.

4 Normalized displacement and cumulative interference

Adjacent alignment concerns one pair of steps. To characterize accumulation over the complete trajectory, we compare the sum of gradient lengths with the length of their sum:

$$A_T := \sum_{t=1}^{T} \|g_t\|, \qquad B_T := \left\| \sum_{t=1}^{T} g_t \right\|. \tag{19}$$

For Eq. (1), ηA_T is the discrete path length and $\eta B_T = \|\Delta_T\|$ is its endpoint displacement. Provided $A_T > 0$, we define

$$\boxed{R_T := \frac{B_T}{A_T} = \frac{\|\Delta_T\|}{\eta \sum_{t=1}^{T} \|g_t\|} \in [0, 1].} \tag{20}$$

The triangle inequality gives this range. Values near one describe coherent accumulation; small values indicate that little of the accumulated step length remains as net parameter motion. A small ratio does not imply small absolute displacement, because the path length in its denominator can be large.

Expanding A_T^2 and B_T^2 makes the connection to pairwise directional disagreement exact:

$$R_T^2 = 1 - \frac{2 \sum_{1 \le u < t \le T} (\|g_u\| \, \|g_t\| - g_u^\top g_t)}{\left(\sum_{t=1}^{T} \|g_t\| \right)^2}. \tag{21}$$

Each difference in the numerator is nonnegative by Cauchy–Schwarz. This metric compares the observed sum with fully aligned accumulation of the same gradient lengths. It includes orthogonal and partially aligned pairs as well as opposing pairs; low R_T does not require every interaction to be negative. Similarly, negative adjacent cosines do not determine the sign of the complete interaction term in Eq. (3).

The normalized displacement measures the geometry of a trajectory, not its progress in objective value. Movement can cancel in a sharply curved direction while continuing to reduce the loss in other directions. The ratio also depends on the selected horizon: a full-prefix measurement includes the earlier aligned phase even after a later phase becomes strongly alternating.

5 Full-batch GD evidence on CIFAR-10

5.1 Experimental setup and measurement

We examine the small CIFAR-10 squared-loss setting used in the released edge-of-stability experiments [1, 2]. Cohen et al. describe a regime in which the maximum training-loss Hessian eigenvalue remains near, and often slightly above, $2/\eta$, while short-term loss increases coexist with longer-term loss reduction [1]. Our experiment asks how the gradient secant and cumulative displacement behave alongside this spectral pattern.

We use the first 5,000 CIFAR-10 training images and a fully connected 3072–200–200–10 network with tanh hidden activations and biases (656,810 parameters). Inputs are centered and standardized per input coordinate using the complete 50,000-image training split. For one-hot targets y_i and network outputs $h_\theta(x_i)$, the objective is

$$\mathcal{L}(\theta) = \frac{1}{2N} \sum_{i=1}^{N} \|h_\theta(x_i) - y_i\|^2, \qquad N = 5000. \tag{22}$$

There is no additional division by the number of classes. Four GD runs share the same frozen initialization from seed zero and use learning rates

$$\eta \in \{0.004, 2/300, 0.01, 0.02\}.$$

Each run completes 20,000 updates with no momentum, weight decay, clipping, augmentation, learning-rate schedule, or injected gradient noise. The full gradient accumulates five physical batches of 1,000 examples before each update. This is full-batch optimization, not mini-batch sampling.

We record loss and raw-gradient diagnostics at every update. The largest algebraic Hessian eigenvalue is estimated on all 5,000 examples at initialization and every 100 updates, including the final state. Each returned eigenpair passes a relative residual check of at most 0.005. We explicitly select the largest algebraic eigenvalue, rather than the largest magnitude; the fixed 20,000-update horizon and frozen initialization are also declared choices of this reproduction. Parameter updates use FP32, with TF32 disabled and FP64 diagnostic reductions. The displacement identities are stated in exact arithmetic; actual parameter displacements are separately audited for rounding.

Figure 1 shows all recorded values without smoothing or temporal thinning. The first adjacent pair is defined at update two; an undefined first value is not replaced by zero. Loss and Hessian values are indexed by parameter state, while the gradient for update t belongs to θ_{t-1}. These labels are retained when comparing trends across the horizon. Hessian measurements are connected for display but remain 100-step samples.

5.2 Alignment reversal and the spectral regime

At $\eta = 0.004$, consecutive gradients remain almost parallel throughout the run. The gradient-derived $\eta\widehat{\kappa}_t$ remains close to zero, whereas $\eta\lambda_{\max}$ increases to approximately 1.68. This separation is consistent with the distinction between gradient-direction curvature and maximum curvature; the secant is not a general estimator of the largest Hessian eigenvalue.

The three larger rates show a different pattern. The first negative adjacent pair occurs at updates 9467, 4565, and 1156 for $\eta = 2/300$, 0.01, and 0.02, respectively. Their late cosine means are approximately -0.999, and their final scaled largest eigenvalues lie between 2.070 and 2.077. Thus, the alignment reversal moves earlier as the learning rate increases within this matched cohort. The loss develops short-term fluctuations but continues to decrease over the fixed horizon.

In this alternating regime, $\eta\widehat{\kappa}_t$ fluctuates near two (Fig. 1c). Equation (14) explains this value directly: if $\rho_t \approx -1$ and consecutive gradient norms are similar, then $1 - \rho_t \|g_t\| / \|g_{t-1}\| \approx 2$. The secant therefore tracks the late edge-of-stability scale in these runs and supplies a directional description of the reversal. It does not follow the entire maximum-eigenvalue curve pointwise. Moreover, the coincidence near two is linked algebraically to the measured reversal; it is not an independent proof of stability.

For a positive quadratic eigendirection of curvature λ, plain GD multiplies that component by $1 - \eta\lambda$ at each step. Alignment reverses when $\eta\lambda > 1$, while contraction requires $0 < \eta\lambda < 2$. The values one and two therefore describe different local questions: directional sign reversal and contraction of a fixed quadratic mode. The nonlinear training trajectory need not obey a frozen-quadratic stability interpretation globally.

Table 2: GD results at $T = 20{,}000$. Bars average the last 1,000 recorded gradient-pair values; loss, maximum curvature and R_T refer to the final state or full prefix. The last recorded gradient belongs to θ_{T-1}; the final loss and Hessian belong to θ_T. One initialization is used.

η	$\mathcal{L}(\theta_T)$	$\overline{\rho}$	$\overline{\eta\widehat{\kappa}}$	$\eta\lambda_{\max}(\theta_T)$	R_T
0.004	0.09688	1.0000	4.35×10^{-5}	1.6767	0.7602
2/300	0.05651	-0.9989	1.9988	2.0762	0.0250
0.01	0.03204	-0.9987	1.9985	2.0702	0.0148
0.02	0.01945	-0.9985	1.9988	2.0773	0.0105

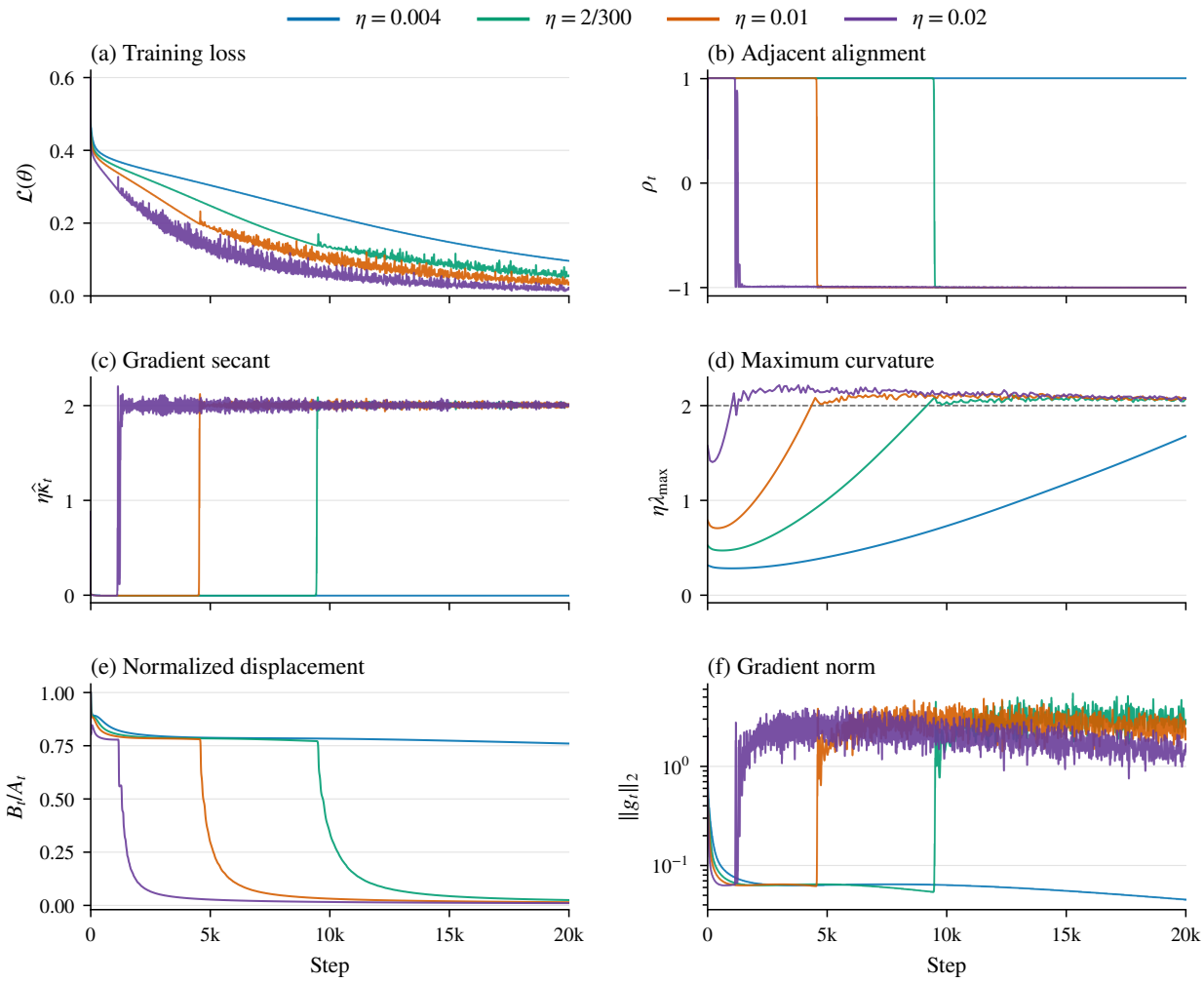


Figure 1: Full-batch GD on CIFAR-10 with four matched learning rates. (a) Training loss; (b) consecutive-gradient cosine; (c) the gradient-derived quantity $\eta\widehat{\kappa}_t = 1-\rho_t \|g_t\| / \|g_{t-1}\|$; (d) the scaled largest algebraic Hessian eigenvalue, with the dashed reference at two; (e) full-prefix normalized displacement B_t/A_t; and (f) raw-gradient norm on log Y. All curves are unsmoothed. Hessians are measured every 100 updates; the other diagnostics are measured every update. Larger learning rates enter the alternating-gradient regime earlier. Loss decreases over the horizon while cumulative cancellation and oscillatory gradient magnitudes increase.

5.3 Cumulative cancellation despite continued loss reduction

The adjacent reversal coincides with a sharp decline in normalized displacement (Fig. 1e). At the horizon, B_T/A_T is 0.7602 for $\eta = 0.004$, but only 0.0250, 0.0148, and 0.0105 at the three larger rates. Hence, only about 2.50%, 1.48%, and 1.05% of the accumulated path length remains as net displacement in these runs. This connects the local reversal diagnostic with substantial cumulative cancellation.

Absolute displacement does not approach zero in this comparison. The audited endpoint distances from initialization are approximately 3.96, 5.09, 5.99, and 7.70, in increasing learning-rate order. At the same time, the corresponding path lengths are approximately 5.21, 203.80, 403.96, and 732.59. The small ratios arise because net motion is small relative to the much longer paths. The complete raw-gradient all-pair interaction also remains positive in all four runs, despite the strongly negative late adjacent pairs at the larger rates. Longer-lag interactions cannot therefore be omitted when interpreting Eq. (5).

Raw-gradient norms fall during the early phase, then increase and fluctuate after the reversal at the larger learning rates (Fig. 1f). Larger step lengths can then contribute little additional net motion because successive directions oppose one another. Oscillation across a curved basin is one possible interpretation, but the scalar diagnostics do not identify a spatial spiral or establish that mechanism. Neither a large gradient norm nor a small displacement ratio alone demonstrates loss instability.

Finally, stronger cancellation does not imply slower convergence in this finite-horizon comparison. The final training loss decreases from 0.09688 at $\eta = 0.004$ to 0.01945 at $\eta = 0.02$. The geometric loss of net path efficiency coexists with greater training-loss reduction. These observations are based on one frozen initialization, a preselected learning-rate grid and a finite horizon; they do not establish a general optimizer ranking or a generalization result.

TODO: an alignment-guided learning-rate intervention. Test a schedule that halves the learning rate when consecutive-gradient cosine becomes negative, specifying the trigger and repeat rule before execution. Compare loss reduction at a matched update budget with fixed-rate controls. The intervention may restore positive alignment without improving convergence; these two outcomes must be evaluated separately.

References

[1] J. M. Cohen, S. Kaur, Y. Li, J. Z. Kolter, and A. Talwalkar. Gradient descent on neural networks typically occurs at the edge of stability. *International Conference on Learning Representations*, 2021. `https://arxiv.org/abs/2103.00065`.

[2] J. M. Cohen and collaborators. Released implementation of the edge-of-stability experiments, `github` branch. `https://github.com/locuslab/edge-of-stability/tree/github`.